

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('default')
sns.set_theme()
```

```
from google.colab import files

df = files.upload()
```

Choose files WA_Fn-Use...-Attrition.csv
WA_Fn-UseC_-HR-Employee-Attrition.csv(text/csv) - 227977 bytes, last modified:
 20/08/2026 - 100% done
 Saving WA_Fn-UseC_-HR-Employee-Attrition.csv to WA_Fn-UseC_-HR-Employee-Attri

```
import pandas as pd

df = pd.read_csv(
    "WA_Fn-UseC_-HR-Employee-Attrition.csv"
)

print(df.shape)
df.head()
```

(1470, 35)

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome
0	41	Yes	Travel_Rarely	1102	Sales	1
1	49	No	Travel_Frequently	279	Research & Development	8
2	37	Yes	Travel_Rarely	1373	Research & Development	2
3	33	No	Travel_Frequently	1392	Research & Development	3
4	27	No	Travel_Rarely	591	Research & Development	2

5 rows × 35 columns

```
df.head()
```

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5 rows × 35 columns

```
print("Rows and Columns:")
print(df.shape)
```

```
Rows and Columns:
(1470, 35)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                   1470 non-null   int64
1   Attrition                           1470 non-null   object
2   BusinessTravel                      1470 non-null   object
3   DailyRate                           1470 non-null   int64
4   Department                          1470 non-null   object
5   DistanceFromHome                    1470 non-null   int64
6   Education                           1470 non-null   int64
7   EducationField                      1470 non-null   object
8   EmployeeCount                       1470 non-null   int64
9   EmployeeNumber                      1470 non-null   int64
10  EnvironmentSatisfaction              1470 non-null   int64
11  Gender                              1470 non-null   object
12  HourlyRate                          1470 non-null   int64
13  JobInvolvement                      1470 non-null   int64
14  JobLevel                            1470 non-null   int64
15  JobRole                             1470 non-null   object
16  JobSatisfaction                     1470 non-null   int64
17  MaritalStatus                      1470 non-null   object
18  MonthlyIncome                       1470 non-null   int64
19  MonthlyRate                         1470 non-null   int64
20  NumCompaniesWorked                  1470 non-null   int64
21  Over18                             1470 non-null   object
22  OverTime                           1470 non-null   object
23  PercentSalaryHike                   1470 non-null   int64
24  PerformanceRating                   1470 non-null   int64
25  RelationshipSatisfaction             1470 non-null   int64
```

```

26 StandardHours      1470 non-null   int64
27 StockOptionLevel   1470 non-null   int64
28 TotalWorkingYears  1470 non-null   int64
29 TrainingTimesLastYear 1470 non-null   int64
30 WorkLifeBalance    1470 non-null   int64
31 YearsAtCompany     1470 non-null   int64
32 YearsInCurrentRole  1470 non-null   int64
33 YearsSinceLastPromotion 1470 non-null   int64
34 YearsWithCurrManager 1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB

```

```
df.describe()
```

	Age	DailyRate	DistanceFromHome	Education	EmployeeCount
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.0
mean	36.923810	802.485714	9.192517	2.912925	1.0
std	9.135373	403.509100	8.106864	1.024165	0.0
min	18.000000	102.000000	1.000000	1.000000	1.0
25%	30.000000	465.000000	2.000000	2.000000	1.0
50%	36.000000	802.000000	7.000000	3.000000	1.0
75%	43.000000	1157.000000	14.000000	4.000000	1.0
max	60.000000	1499.000000	29.000000	5.000000	1.0

8 rows × 26 columns

```
print(df.columns.tolist())
```

```
['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department', 'DistanceFr
```

```
df.isnull().sum()
```


	0
Age	0
Attrition	0
BusinessTravel	0

```
df['Attrition'].value_counts()
```

Department	count	0
DistanceFromHome	0	
Attrition	0	
Education	0	
No	1233	0
EducationField	0	
Yes	237	0
EmployeeCount	0	
EmployeeNumber	0	

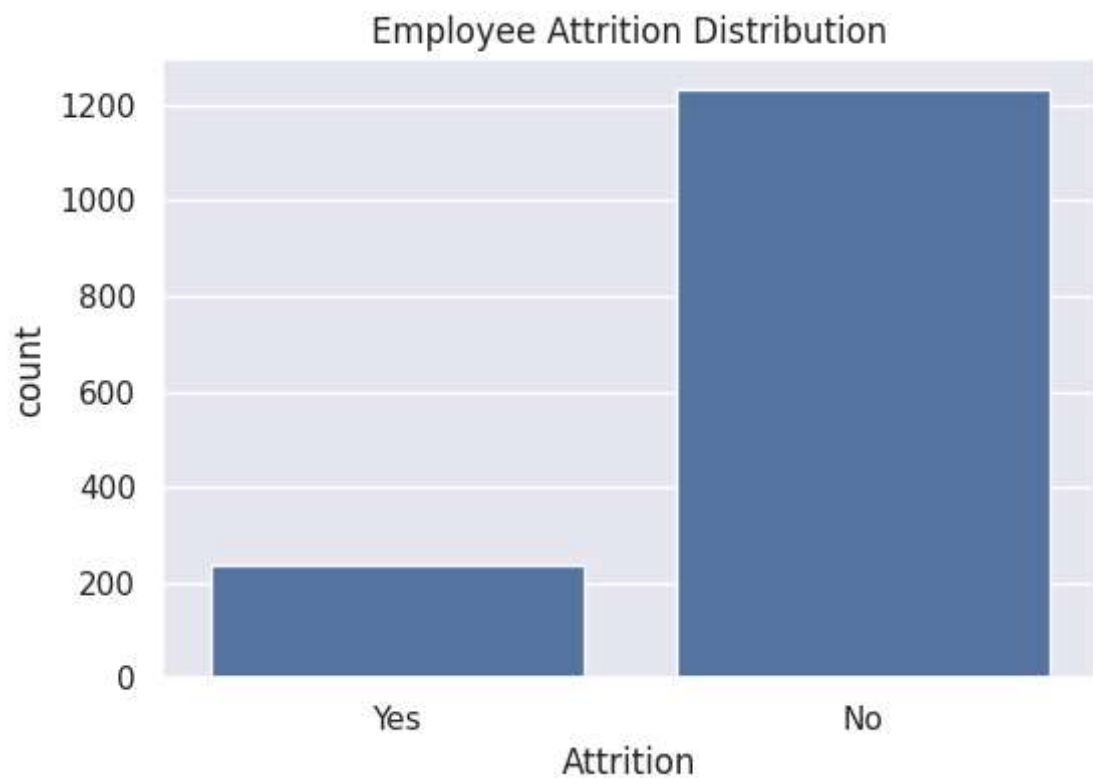
dtype: int64

```
plt.figure(figsize=(6,4))

sns.countplot(
    x='Attrition',
    data=df
)

plt.title("Employee Attrition Distribution")

plt.show()
```



WorkLifeBalance	0
-----------------	---

```
plt.figure(figsize=(8,5))

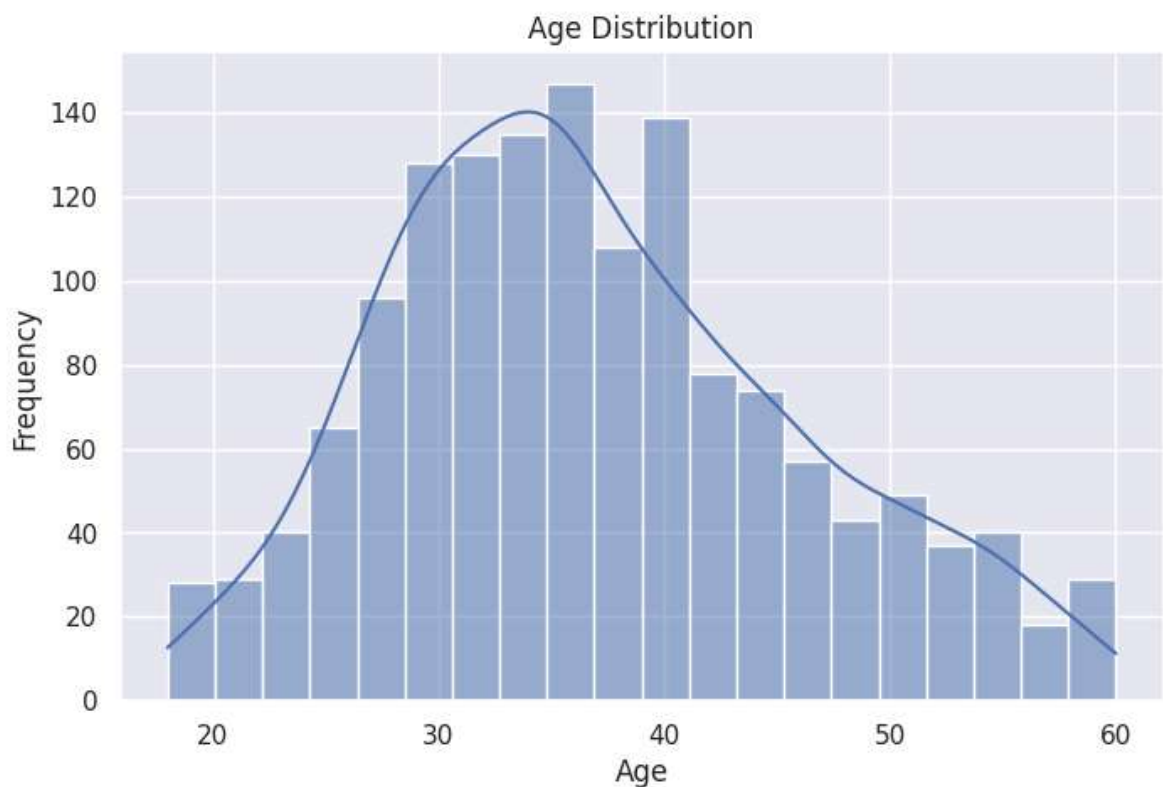
sns.histplot(
    df['Age'],
    bins=20,
    kde=True
)

plt.title("Age Distribution")

plt.xlabel("Age")

plt.ylabel("Frequency")

plt.show()
```

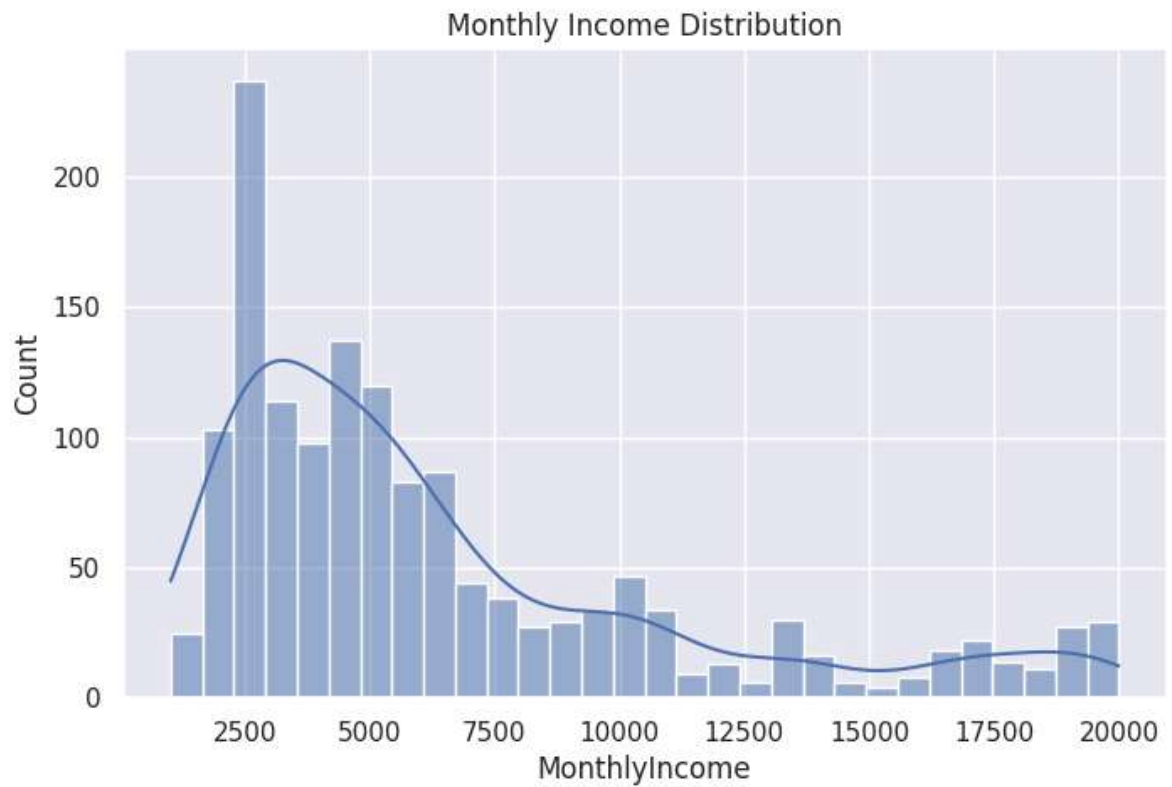


```
plt.figure(figsize=(8,5))

sns.histplot(
    df['MonthlyIncome'],
    bins=30,
    kde=True
)

plt.title("Monthly Income Distribution")

plt.show()
```

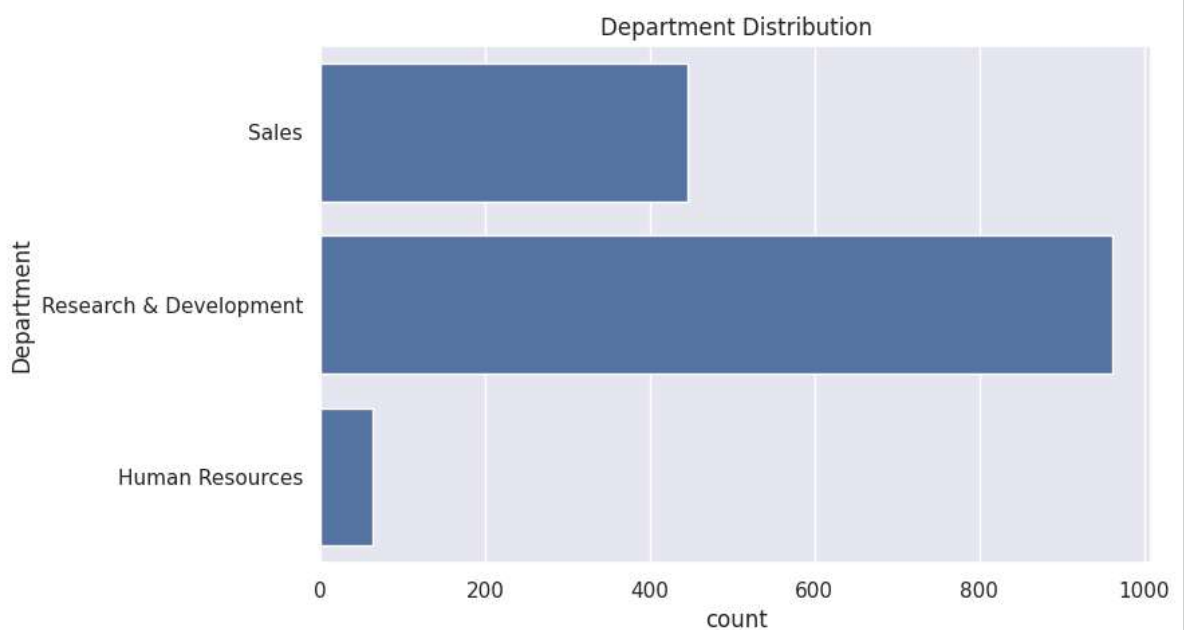


```
plt.figure(figsize=(8,5))

sns.countplot(
    y='Department',
    data=df
)

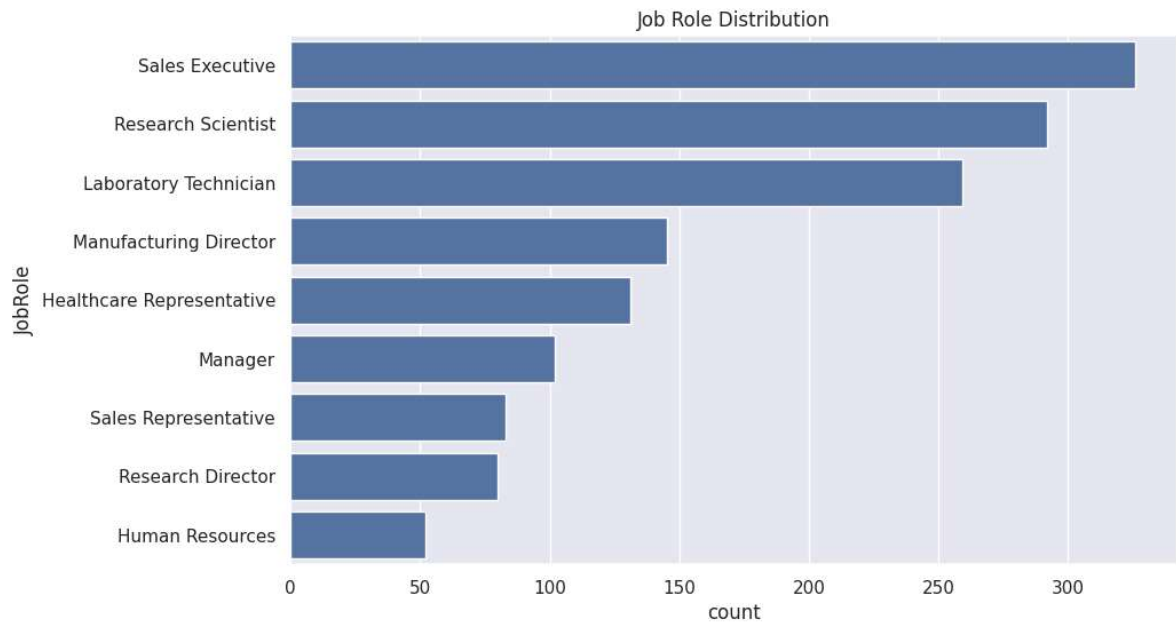
plt.title("Department Distribution")

plt.show()
```

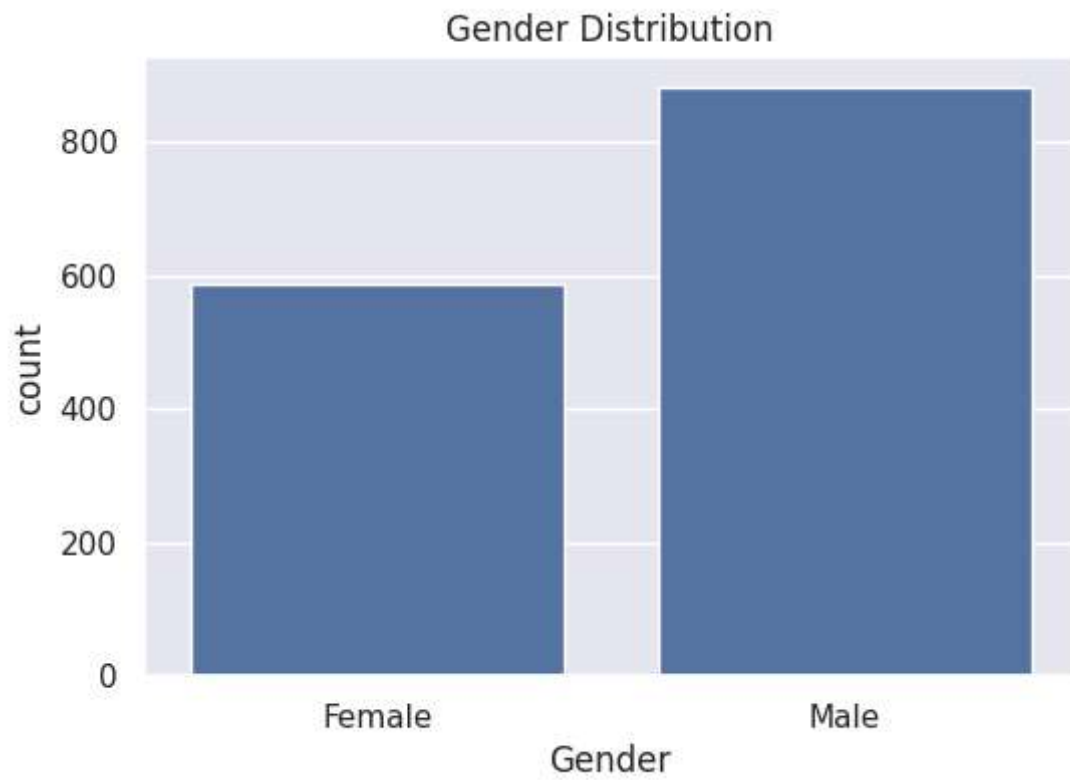


```
plt.figure(figsize=(10,6))
```

```
sns.countplot(  
    y='JobRole',  
    data=df  
)  
  
plt.title("Job Role Distribution")  
  
plt.show()
```



```
plt.figure(figsize=(6,4))  
  
sns.countplot(  
    x='Gender',  
    data=df  
)  
  
plt.title("Gender Distribution")  
  
plt.show()
```

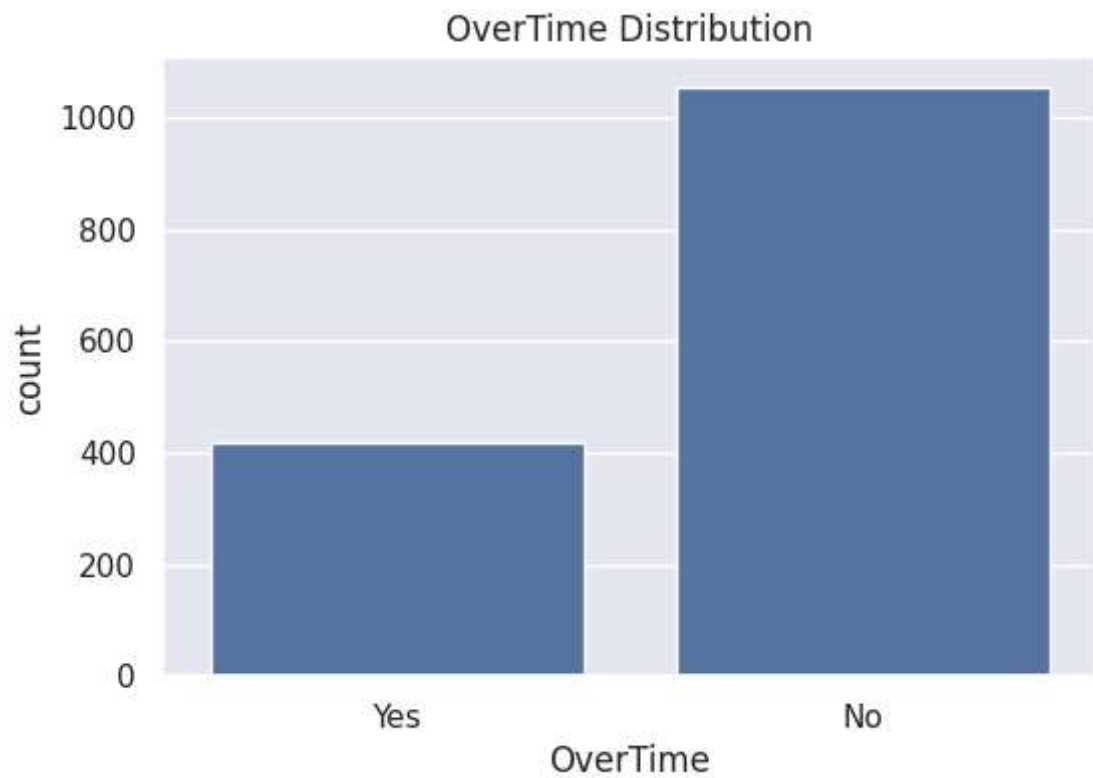


```
plt.figure(figsize=(6,4))

sns.countplot(
    x='OverTime',
    data=df
)

plt.title("OverTime Distribution")

plt.show()
```



```
from sklearn.preprocessing import LabelEncoder

temp_df = df.copy()

encoder = LabelEncoder()

for col in temp_df.select_dtypes(include='object'):
    temp_df[col] = encoder.fit_transform(temp_df[col])

print("Encoding Completed")
```

Encoding Completed

```
corr_matrix = temp_df.corr()

corr_matrix.head()
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome
Age	1.000000	-0.159205	0.024751	0.010661	-0.031882	0.000000
Attrition	-0.159205	1.000000	0.000074	-0.056652	0.063991	0.000000
BusinessTravel	0.024751	0.000074	1.000000	-0.004086	-0.009044	0.000000
DailyRate	0.010661	-0.056652	-0.004086	1.000000	0.007109	0.000000
Department	-0.031882	0.063991	-0.009044	0.007109	1.000000	0.000000

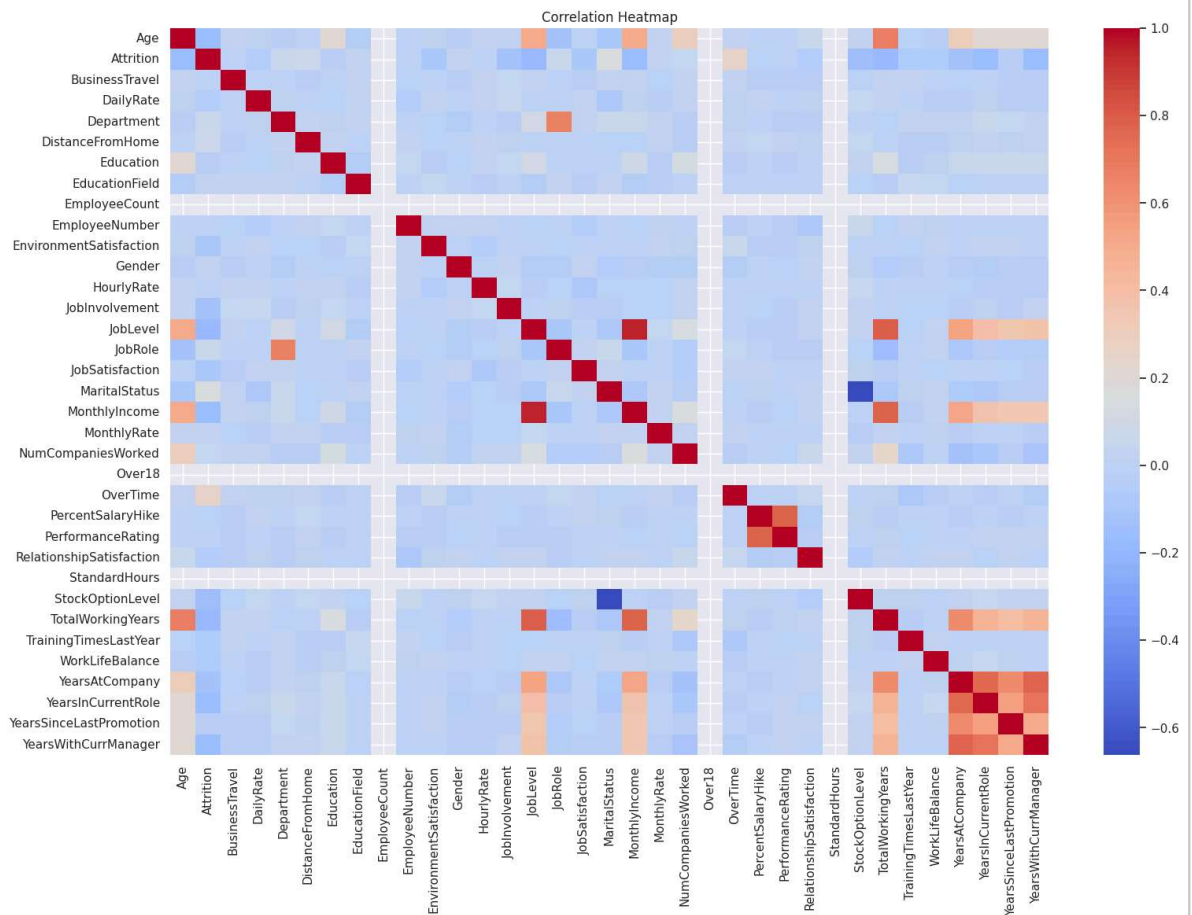
5 rows × 35 columns

```
plt.figure(figsize=(18,12))

sns.heatmap(
    corr_matrix,
    cmap='coolwarm'
)

plt.title("Correlation Heatmap")

plt.show()
```

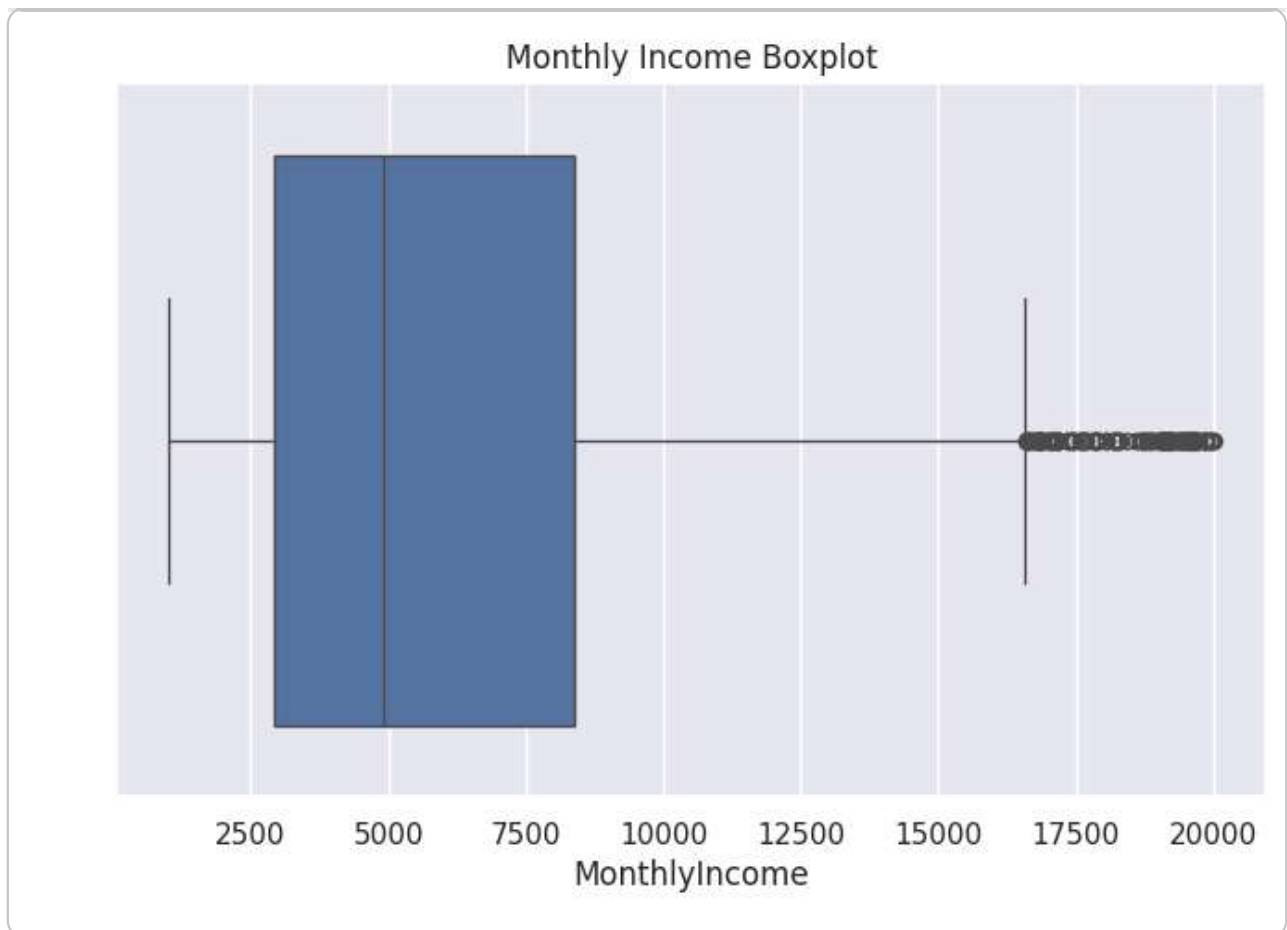


```
plt.figure(figsize=(8,5))

sns.boxplot(
    x=df['MonthlyIncome']
)

plt.title("Monthly Income Boxplot")

plt.show()
```

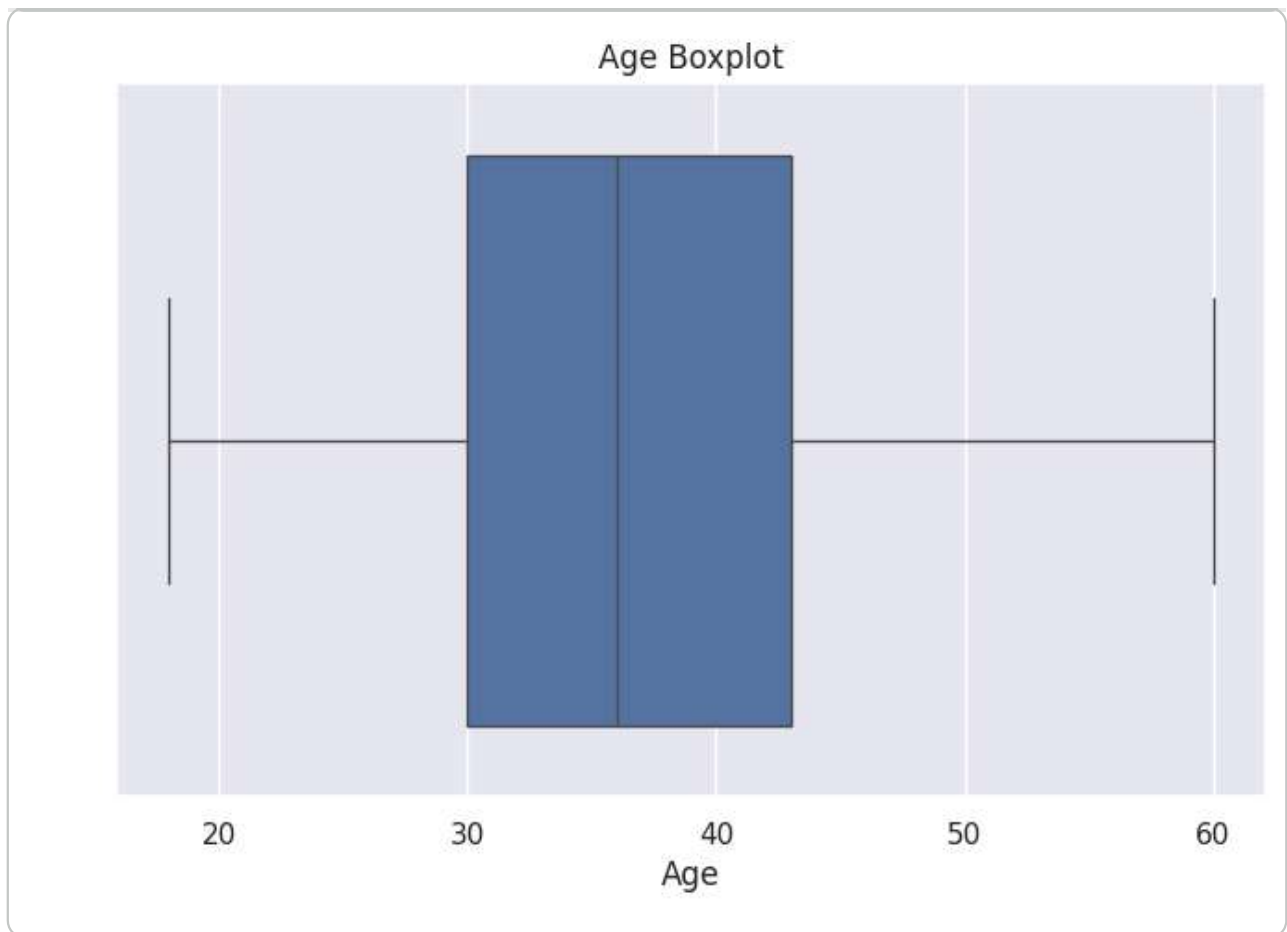


```
plt.figure(figsize=(8,5))

sns.boxplot(
    x=df['Age']
)

plt.title("Age Boxplot")

plt.show()
```

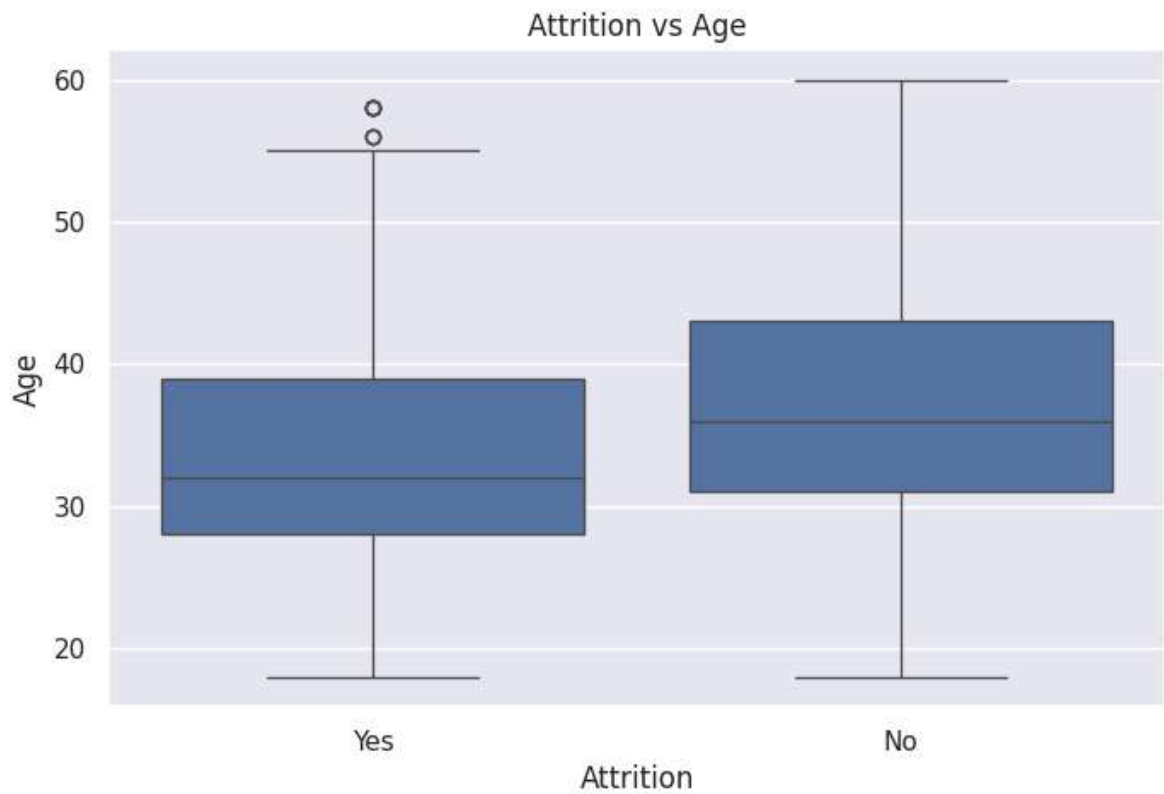


```
plt.figure(figsize=(8,5))

sns.boxplot(
    x='Attrition',
    y='Age',
    data=df
)

plt.title("Attrition vs Age")

plt.show()
```

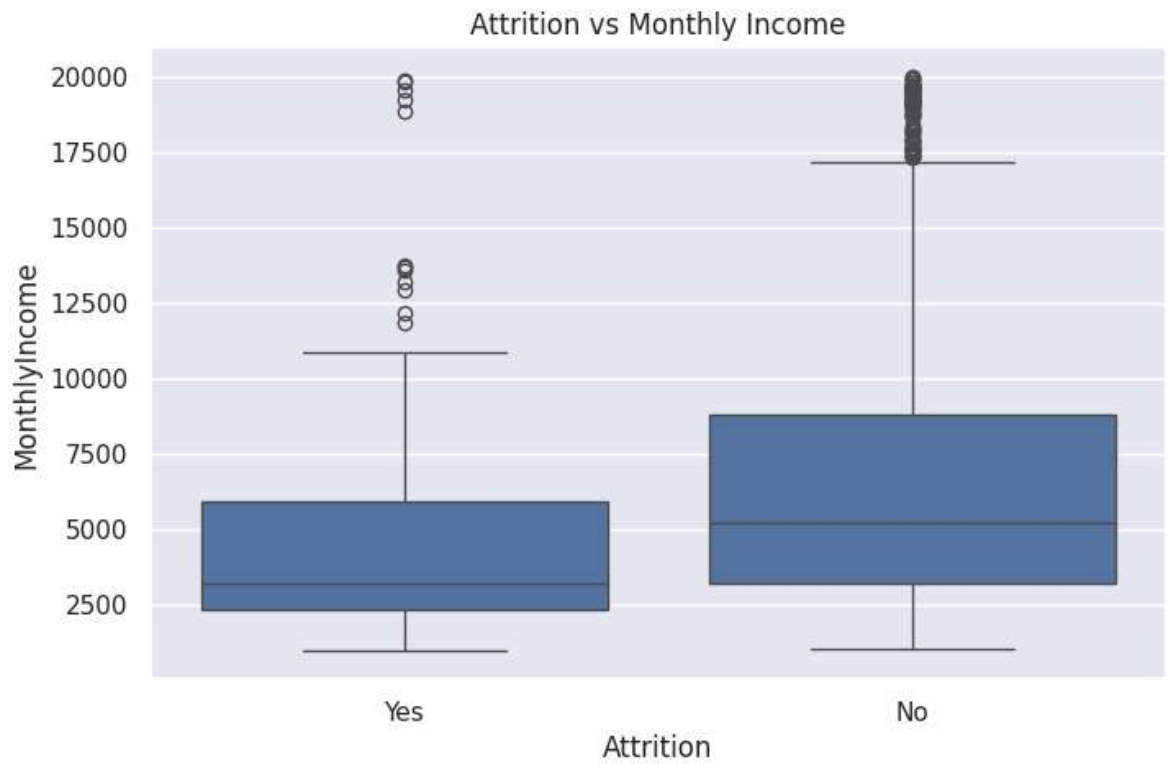


```
plt.figure(figsize=(8,5))

sns.boxplot(
    x='Attrition',
    y='MonthlyIncome',
    data=df
)

plt.title("Attrition vs Monthly Income")

plt.show()
```

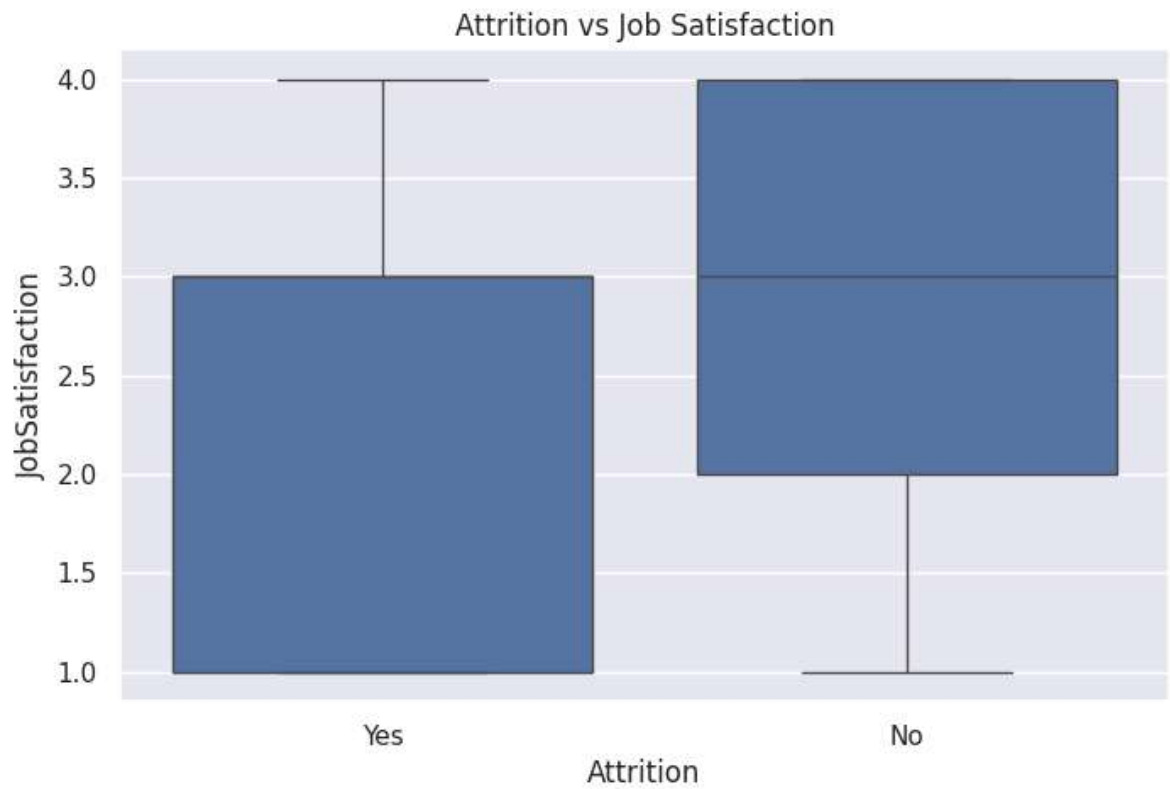


```
plt.figure(figsize=(8,5))

sns.boxplot(
    x='Attrition',
    y='JobSatisfaction',
    data=df
)

plt.title("Attrition vs Job Satisfaction")

plt.show()
```



```
plt.figure(figsize=(8,5))

sns.scatterplot(
    x='Age',
    y='MonthlyIncome',
    hue='Attrition',
    data=df
)

plt.title("Age vs Monthly Income")

plt.show()
```

Age vs Monthly Income

20000

Attrition



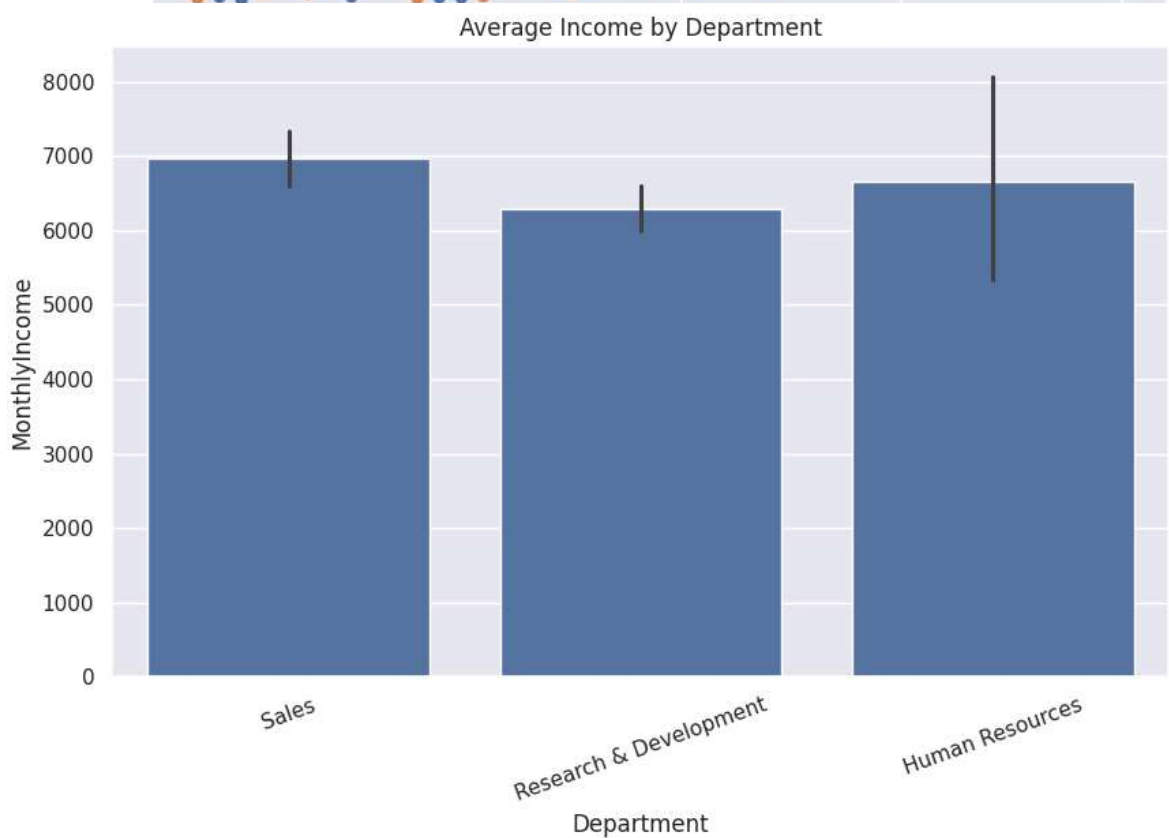
```
plt.figure(figsize=(10,6))

sns.barplot(
    x='Department',
    y='MonthlyIncome',
    data=df
)

plt.title("Average Income by Department")

plt.xticks(rotation=20)

plt.show()
```



```
plt.figure(figsize=(10,6))

sns.countplot(
    x='Department',
    hue='Attrition',
    data=df
)
```